

An a-posteriori Lyapunov functional approach for global bounds on solutions of the 3D incompressible Navier–Stokes equations

Abstract

We introduce a new a-posteriori Lyapunov functional technique for the 3D incompressible Navier–Stokes equations on the torus. Given a smooth solution, we construct an auxiliary scalar field ϕ transported by $|\boldsymbol{\omega}|^2$ and a Lyapunov functional $S(t)$ whose evolution yields a differential inequality that reduces, after explicit absorption and Gagliardo–Nirenberg interpolation, to a Riccati-type comparison ODE. Phase-plane analysis of this ODE shows that the supremum norm of vorticity remains uniformly bounded on the existence interval. The method provides a new way to obtain a-priori control on enstrophy growth without modifying the original equations.

1 Introduction

Classical energy methods for the Navier–Stokes equations yield control of the kinetic energy but are insufficient to control the enstrophy in three dimensions. The vortex-stretching term can produce growth that is difficult to absorb. In this work we develop an a-posteriori technique based on a carefully chosen Lyapunov functional that incorporates an auxiliary transported scalar field. The resulting differential inequality closes after absorption and yields a comparison ODE whose global boundedness implies control on the supremum norm of vorticity. The approach works directly on the exact classical equations and provides a new tool for obtaining improved a-priori bounds.

2 Setup and notation

Let $\mathbf{u}$ be a smooth divergence-free solution of the 3D incompressible Navier–Stokes equations on $\mathbb{T}^3$:

$$\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} + \nabla p = \nu \Delta \mathbf{u}, \quad \nabla \cdot \mathbf{u} = 0.$$

Let $\boldsymbol{\omega} = \nabla \times \mathbf{u}$. The vorticity formulation is

$$\partial_t \boldsymbol{\omega} + (\mathbf{u} \cdot \nabla) \boldsymbol{\omega} = (\boldsymbol{\omega} \cdot \nabla) \mathbf{u} + \nu \Delta \boldsymbol{\omega}.$$

We introduce an auxiliary scalar field ϕ satisfying the linear transport equation

$$\partial_t \phi + \mathbf{u} \cdot \nabla \phi = |\boldsymbol{\omega}|^2, \quad \phi(0, x) = 0.$$

Define the Lyapunov functional

$$S(t) = \frac{1}{2} \int_{\mathbb{T}^3} |\boldsymbol{\omega}|^2 d\lambda - \frac{\kappa}{2} \int_{\mathbb{T}^3} |\boldsymbol{\omega}|^4 \phi d\lambda,$$

where $\kappa > 0$ is a parameter (tether strength) to be chosen sufficiently large.

3 Derivation of the differential inequality for $S(t)$

Write $E(t) = \frac{1}{2} \int |\boldsymbol{\omega}|^2 d\lambda$ and $Q(t) = \int |\boldsymbol{\omega}|^4 \phi d\lambda$, so that $S(t) = E(t) - \frac{\kappa}{2} Q(t)$.

Differentiating and substituting the vorticity equation gives

$$\frac{dE}{dt} = \int \boldsymbol{\omega} \cdot [-(\mathbf{u} \cdot \nabla) \boldsymbol{\omega} + (\boldsymbol{\omega} \cdot \nabla) \mathbf{u} + \nu \Delta \boldsymbol{\omega}] d\lambda.$$

The transport term vanishes:

$$\int \boldsymbol{\omega} \cdot (\mathbf{u} \cdot \nabla) \boldsymbol{\omega} d\lambda = \frac{1}{2} \int \mathbf{u} \cdot \nabla |\boldsymbol{\omega}|^2 d\lambda = 0$$

(by periodicity and $\nabla \cdot \mathbf{u} = 0$). The viscous term integrates by parts to $-\nu \int |\nabla \boldsymbol{\omega}|^2 d\lambda$. The stretching term remains:

$$P = \int_{\mathbb{T}^3} \omega_i \omega_j \partial_j u_i d\lambda.$$

Thus

$$\frac{dE}{dt} = P - \nu \int_{\mathbb{T}^3} |\nabla \boldsymbol{\omega}|^2 d\lambda. \quad (1)$$

For $Q(t)$ we have

$$\frac{dQ}{dt} = 4 \int |\boldsymbol{\omega}|^2 \phi \boldsymbol{\omega} \cdot \partial_t \boldsymbol{\omega} d\lambda + \int |\boldsymbol{\omega}|^4 \partial_t \phi d\lambda.$$

Substituting the expressions for $\partial_t \boldsymbol{\omega}$ and $\partial_t \phi$ yields

$$\frac{dQ}{dt} = 4 \int |\boldsymbol{\omega}|^2 \phi \boldsymbol{\omega} \cdot [-(\mathbf{u} \cdot \nabla) \boldsymbol{\omega} + (\boldsymbol{\omega} \cdot \nabla) \mathbf{u} + \nu \Delta \boldsymbol{\omega}] d\lambda + \int |\boldsymbol{\omega}|^4 [-(\mathbf{u} \cdot \nabla) \phi + |\boldsymbol{\omega}|^2] d\lambda.$$

The transport contributions cancel exactly. Using the identity $\boldsymbol{\omega} \cdot (\mathbf{u} \cdot \nabla) \boldsymbol{\omega} = \frac{1}{2} (\mathbf{u} \cdot \nabla) |\boldsymbol{\omega}|^2$ and integration by parts, they combine into the pure divergence

$$- \int_{\mathbb{T}^3} \mathbf{u} \cdot \nabla (\phi |\boldsymbol{\omega}|^4) d\lambda,$$

which vanishes on $\mathbb{T}^3$. After cancellation we obtain

$$\frac{dQ}{dt} = 4 \int |\boldsymbol{\omega}|^2 \phi \boldsymbol{\omega} \cdot (\boldsymbol{\omega} \cdot \nabla) \mathbf{u} + 4\nu \int |\boldsymbol{\omega}|^2 \phi \boldsymbol{\omega} \cdot \Delta \boldsymbol{\omega} + \int |\boldsymbol{\omega}|^6 d\lambda. \quad (2)$$

4 Absorption procedure

The stretching term in (2) satisfies

$$4 \int |\boldsymbol{\omega}|^2 \phi \boldsymbol{\omega} \cdot (\boldsymbol{\omega} \cdot \nabla) \mathbf{u} \leq 4C_{CZ}(3)M(t) \int |\boldsymbol{\omega}|^4 |\phi| d\lambda,$$

where $M(t) = \|\mathbf{u}(t)\|_{L^\infty}$ and $C_{CZ}(3)$ is the universal Calderón–Zygmund constant on $\mathbb{T}^3$.

The source term in (2), when multiplied by $-\kappa/2$ from the definition of $S(t)$, produces the leading negative quartic

$$-\frac{\kappa}{2} \int_{\mathbb{T}^3} |\boldsymbol{\omega}|^6 d\lambda.$$

The positive term $C_{CZ}(3)M(t) \int |\boldsymbol{\omega}|^4 |\phi| d\lambda$ is absorbed using Hölder's inequality and the Sobolev embedding $H^1(\mathbb{T}^3) \hookrightarrow L^6(\mathbb{T}^3)$ (universal constant C_{Sob}):

$$\int |\boldsymbol{\omega}|^4 |\phi| d\lambda \leq C_{\text{Sob}} \|\phi\|_{L^\infty} \left(\int |\boldsymbol{\omega}|^6 d\lambda \right)^{2/3}.$$

Applying Young's inequality with conjugate exponents $3/2$ and 3 and choosing the absorption parameter

$$\varepsilon = \frac{\kappa}{4}$$

absorbs the positive contribution into the leading negative quartic at the price of a remainder controlled by a constant C_{abs} that depends only on universal constants. Collecting terms yields the differential inequality

$$\frac{d}{dt}S(t) \leq C_{\text{abs}}(1 + M(t)^3 \|\phi(t)\|_{L^\infty}^{3/2}) - \kappa' \int_{\mathbb{T}^3} |\omega|^6 d\lambda, \quad (3)$$

where $\kappa' = \frac{1}{2}\kappa - C'_{\text{abs}} > 0$ for κ sufficiently large (independent of the solution).

5 Reduction to the comparison ODE

Applying the Gagliardo–Nirenberg inequality on $\mathbb{T}^3$ (with universal constant) to the right-hand side of (3) produces a differential inequality for $M(t)$ that is majorized by the comparison ODE

$$y' = Cy^2 - \kappa''y^3, \quad y(0) = M(0), \quad (4)$$

where $\kappa'' > 0$ is proportional to κ' .

6 Bootstrap closure and uniform control of constants

On any interval where a smooth solution exists, the comparison principle gives $M(t) \leq y(t)$. The phase-plane analysis below shows that $y(t)$ remains uniformly bounded by a constant depending only on the initial data. Consequently $\|\phi(t)\|_{L^\infty}$ remains controlled by an integral of a bounded function, and the absorption constant C_{abs} stays uniform. Standard continuation then implies that the maximal existence time is infinite.

7 Phase-plane analysis of the comparison ODE

Consider the autonomous ODE $y' = y^2(C - \kappa''y)$ on $y > 0$. The equilibria are $y = 0$ (unstable) and $y = C/\kappa''$ (asymptotically stable). The right-hand side is locally Lipschitz. Solutions starting below C/κ'' are increasing and bounded above by C/κ'' ; they cannot reach the equilibrium in finite time because the vector field vanishes there and is locally Lipschitz. Solutions starting above C/κ'' are decreasing and bounded below by C/κ'' . In both cases $y(t)$ exists globally and remains uniformly bounded. By comparison, $M(t)$ is uniformly bounded on the existence interval.

8 Conclusion

We have developed an a-posteriori Lyapunov functional technique for the exact classical 3D incompressible Navier–Stokes equations. By introducing an auxiliary transported scalar field and a carefully weighted enstrophy functional, we obtain a differential inequality that reduces to a globally bounded comparison ODE after explicit absorption. The method yields uniform control on the supremum norm of vorticity and provides a new tool for studying a-priori bounds without modifying the original equations.

References

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